

Predicting Vehicle Fuel Efficiency Using Linear Regression

An investigation of the Auto MPG dataset

This report investigates how accurately linear regression can predict vehicle fuel efficiency in the Auto MPG dataset, and whether adding horsepower and model year improves predictions compared with using vehicle weight alone. The investigation explores the relationship between vehicle characteristics and fuel efficiency while assessing the usefulness and limitations of simple predictive models.

Executive Summary

This investigation examines how accurately linear regression can predict vehicle fuel efficiency in the Auto MPG dataset, and whether horsepower and model year improve predictions beyond vehicle weight alone.

After removing six observations with missing horsepower values, 392 complete observations remained. Three approaches were compared: a baseline predicting the training mean, a weight-only linear regression, and a multiple linear regression using weight, horsepower and model year. Evaluation used five-fold cross-validation within the training set and a separate test set of 79 observations.

Weight-only regression reduced test mean absolute error from 5.881 to 3.464 miles per gallon, an improvement of 41.1% over the baseline. Multiple regression reduced this error further to 2.509 miles per gallon, improving on weight-only regression by 27.6%. It also achieved the lowest mean cross-validation error, at 2.695 miles per gallon.

These results indicate that horsepower and model year jointly add predictive value beyond weight. However, systematic residual patterns remained, suggesting that the linear models did not capture all relevant variation. The evaluation concerns vehicles from model years 1970–1982 and does not establish accuracy for modern vehicles. Exploratory analysis also used the full dataset, so any resulting influence on predictor selection limits the independence of the test evaluation.

Main takeaway

Adding horsepower and model year to weight reduced test mean absolute error by 27.6%, from 3.464 to 2.509 miles per gallon. The multiple regression model performed best among the three approaches evaluated on this historical dataset.

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1 Introduction

A vehicle's fuel efficiency describes the distance it can travel on a given amount of fuel. In this dataset, it is measured in miles per gallon (MPG), with higher values indicating greater fuel efficiency.

In this investigation, I examine how vehicle weight, horsepower and model year relate to fuel efficiency. I compare a regression model using weight alone with one that also includes horsepower and model year, to assess whether the additional information improves predictions.

1.1 Research question and objectives

Research question: How accurately can linear regression predict fuel efficiency in the Auto MPG dataset, and does including horsepower and model year improve predictions compared with using vehicle weight alone?

- Prepare and explore the Auto MPG dataset, checking data quality and examining relationships between vehicle characteristics and fuel efficiency, measured in miles per gallon.
- Fit a linear regression model using vehicle weight alone and a multiple linear regression model using weight, horsepower and model year. Compare both with a baseline that predicts the average fuel efficiency of vehicles in the training data.
- Evaluate predictive performance using five-fold cross-validation on the training data. Measure performance on the test data using metrics such as mean absolute error, root mean squared error and the coefficient of determination (R^2). The residual plots are examined as well to identify patterns in prediction errors.

1.2 Scope

This investigation covers data cleaning, exploratory analysis, model fitting and evaluation using the Auto MPG dataset. It compares two linear regression models with a baseline that predicts the mean fuel efficiency of the training data.

I chose this project to put the data-analysis methods I have studied into practice and gain experience explaining the results in a report. The analysis is limited to the historical vehicles in the dataset. Regularised and nonlinear models are outside its scope, but could be explored in future work.

2 Data or Materials

This investigation uses the Auto MPG dataset from the University of California, Irvine Machine Learning Repository [1]. The input dataset contains 398 observations and nine variables, covering vehicles from model years 1970 to 1982. The outcome of interest is city-cycle fuel efficiency, measured in miles per gallon.

The recorded vehicle characteristics are cylinder count, engine displacement, horsepower, weight, acceleration, model year and origin. A vehicle name is also included. The regression models in this investigation use weight, measured in pounds; engine power, measured in horsepower; and model year, recorded as a two-digit value.

The dataset provides historical observations for exploring relationships and evaluating predictions.

Its coverage does not support assuming that the findings apply to modern vehicles.

2.1 Preparation and quality checks

The raw data were loaded using the pandas library in Python. Column names were assigned during import, and question marks were interpreted as missing values. Initial checks examined the dataset dimensions, column types, missing values, duplicate rows and numerical summary statistics.

Six observations contained missing horsepower values. The cleaning procedure removed duplicate rows, if present, and excluded observations with any missing values. This produced a dataset of 392 complete observations, which was saved as a separate file for exploratory analysis and modelling. Missing values were not estimated or replaced, and the original measurement scales were retained.

The same cleaned dataset was used for both regression models, allowing their performance to be compared using the same observations. This also excluded the six incomplete observations from the weight-only model, although horsepower was not required by that model. Removing these observations simplified the comparison but could introduce bias if the excluded vehicles differed systematically from those retained.

Table 1. Outcome and predictors used in the regression models.

Variable	Description	Unit
Fuel efficiency	City-cycle fuel efficiency; the outcome predicted by both regression models.	Miles per gallon
Weight	Vehicle weight; a predictor in both regression models.	Pounds
Horsepower	Engine power; an additional predictor in the multiple regression model.	Horsepower
Model year	Vehicle model year, recorded from 70 to 82 to represent 1970 to 1982; an additional predictor in the multiple regression model.	Year (two-digit code)

3 Methods

The analysis used the 392 complete observations obtained after cleaning. Fuel efficiency was treated as a continuous outcome. Three approaches were compared: a baseline predicting the mean fuel efficiency of the training observations, a simple linear regression using vehicle weight, and a multiple linear regression using weight, horsepower and model year.

The data were divided into training and test sets using an 80:20 split with a random seed of 42. This produced 313 training observations and 79 test observations. Model coefficients were estimated using the training observations only. Five-fold cross-validation within the training set was used to assess predictive performance, followed by evaluation on the test set.

3.1 Reproducibility

The analysis was implemented in Python using pandas for data handling, Matplotlib and Seaborn for visualisation, and scikit-learn for regression modelling and evaluation. The code and data are available in the project repository: <https://github.com/joshmacd/Regression-Models>.

The workflow comprises four scripts, executed in the following order:

1. `01_clean_data.py`: inspect and clean the data.
2. `02_eda.py`: explore the data and generate figures.
3. `03_simple_linear_regression.py`: fit and evaluate the weight-only model and mean baseline.
4. `04_multiple_linear_regression.py`: fit the multiple regression model and compare all three approaches.

These scripts are located in `scripts/auto_mpg/` and should be run from the repository's root directory.

The training and test split used `test_size=0.20` and `random_state=42`. Cross-validation used five folds with shuffling enabled and `random_state=42`. Both regression models included an intercept and used unscaled predictors. The baseline used `DummyRegressor(strategy="mean")`, and cross-validation used `scoring="neg_mean_absolute_error"`. The returned scores were multiplied by -1 to report positive mean absolute errors.

The results reported here were generated the software versions listed in Table 2.

Table 2. Software versions used to generate the reported results.

Software	Version
Python	3.12.1
pandas	3.0.5
Matplotlib	3.11.1
Seaborn	0.13.2
scikit-learn	1.9.0
NumPy	2.5.2
SciPy	1.18.1

3.2 Model or analytical approach

I have included the following derivations for completeness and because I enjoy the mathematics behind the methods. Readers interested mainly in the findings can skip the proofs without losing the main argument.

To begin, let $y_i \in \mathbb{R}$ denote the observed fuel efficiency of vehicle i , and let $w_i, h_i \in \mathbb{R}$ and $t_i \in \mathbb{N}$ denote its weight, horsepower and model year, respectively. The two regression specifications were:

$$y_i = \beta_0 + \beta_1 w_i + \varepsilon_i, \quad (1)$$

$$y_i = \beta_0 + \beta_1 w_i + \beta_2 h_i + \beta_3 t_i + \varepsilon_i, \quad (2)$$

where ε_i represents variation not captured by the specified linear relationship. The coefficients are estimated separately for each model.

In the multiple regression model, each slope describes the change in predicted fuel efficiency associated with a one-unit increase in its predictor, holding the other predictors fixed. These coefficients describe associations and are not interpreted as causal effects. The intercept corresponds to all predictors being zero, which lies outside the observed vehicle characteristics and therefore has no useful physical interpretation here.

3.2.1 Least-squares estimation, [2].

For either regression model, let n be the number of training observations and p the number of predictors. Define the response vector and design matrix by

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n, \quad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} \in \mathbb{R}^{n \times (p+1)}.$$

The first column of $\mathbf{X}$ accounts for the intercept. Ordinary least squares estimates the coefficient vector by minimising the residual sum of squares:

$$\hat{\boldsymbol{\beta}} \in \arg \min_{\mathbf{b} \in \mathbb{R}^{p+1}} S(\mathbf{b}), \quad S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2. \quad (3)$$

Proposition. If $\mathbf{X}$ has linearly independent columns, the unique least-squares solution is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (4)$$

Proof. Expanding the objective gives

$$S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2 = (\mathbf{y} - \mathbf{X}\mathbf{b})^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b}.$$

Differentiation with respect to $\mathbf{b}$ yields

$$\frac{dS(\mathbf{b})}{d\mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b}.$$

A stationary point therefore satisfies the following

$$\frac{dS(\mathbf{b})}{d\mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b} = 0.$$

Hence we have

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}.$$

For every nonzero vector $\mathbf{v}$, full column rank implies that $\mathbf{X}\mathbf{v} \neq 0$. By positive definiteness of the Euclidean norm,

$$\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \|\mathbf{X}\mathbf{v}\|_2^2 > 0.$$

Thus $\mathbf{X}^\top \mathbf{X}$ is positive definite and hence invertible. The Hessian of S is

$$2\mathbf{X}^\top \mathbf{X},$$

which is also positive definite, so the stationary point is the unique global minimum. $\square$

Equation 4 is a theoretical expression. The models were fitted using the `LinearRegression`

estimator in scikit-learn, rather than explicitly calculating the matrix inverse. Full column rank is the condition for the uniqueness result above; it is not established by the proof for this particular dataset.

Predictions for an observation with predictor vector $\mathbf{x}_i$ are consequently

$$\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_{ij}.$$

3.2.2 Mean-prediction baseline

The baseline predicts the same value for every vehicle: the mean fuel efficiency of the training observations,

$$\bar{y}_{\text{train}} = \frac{1}{n} \sum_{i=1}^n y_i.$$

This is the constant prediction that minimises squared error on the training data.

Proof - [3, Section 2.1, Theorem 1(a)]. For any constant $c \in \mathbb{R}$, we see that $y_i - c = y_i - \bar{y}_{\text{train}} + \bar{y}_{\text{train}} - c$ and hence

$$\sum_{i=1}^n (y_i - c)^2 = \sum_{i=1}^n (y_i - \bar{y}_{\text{train}})^2 + n(c - \bar{y}_{\text{train}})^2.$$

Since

$$\sum_{i=1}^n (y_i - \bar{y}_{\text{train}}) = \sum_{i=1}^n y_i - n\bar{y}_{\text{train}} = 0,$$

the cross-term in the expansion vanishes. The first term, $\sum_{i=1}^n (y_i - \bar{y}_{\text{train}})^2$, depends only on the observed training data and is therefore constant with respect to c . For every $c \in \mathbb{R}$,

$$(c - \bar{y}_{\text{train}})^2 \geq 0.$$

Since $n > 0$, it follows that

$$n(c - \bar{y}_{\text{train}})^2 \geq 0.$$

Hence,

$$\sum_{i=1}^n (y_i - c)^2 = \sum_{i=1}^n (y_i - \bar{y}_{\text{train}})^2 + n(c - \bar{y}_{\text{train}})^2 \geq \sum_{i=1}^n (y_i - \bar{y}_{\text{train}})^2.$$

This establishes a lower bound on the sum of squared errors for any constant prediction c .

The bound is attained if and only if

$$n(c - \bar{y}_{\text{train}})^2 = 0 \iff c = \bar{y}_{\text{train}}.$$

Thus the training mean attains the lower bound, and every other constant prediction produces a strictly larger sum of squared errors. Therefore, $c = \bar{y}_{\text{train}}$ is the unique minimiser. $\square$

This baseline measures the benefit of using vehicle characteristics rather than a single constant prediction. Its optimality concerns training squared error; it does not imply optimality for absolute error or test-set performance.

3.2.3 Why evaluation requires unseen observations

A model's ability to fit the training observations does not, by itself, establish its ability to predict fuel efficiency for other vehicles. Additional predictors may capture useful relationships but they may also fit random variation specific to the training sample, this behaviour is known as overfitting, [5, Section 2.2].

The weight-only model is contained within the multiple regression model. We have that, setting the horsepower and model-year coefficients to zero gives the following

$$\beta_0 + \beta_1 w_i + 0h_i + 0t_i = \beta_0 + \beta_1 w_i.$$

Thus every prediction for the weight-only model is also available to the multiple regression model, [4].

Proposition. When both models are fitted by ordinary least squares to the same training observations, the minimum training sum of squared errors for the multiple regression model cannot exceed that of the weight-only model.

Proof. Define the two training sum of squared errors as

$$\begin{aligned} S_{\text{weight}}(a_0, a_1) &= \sum_{i=1}^n (y_i - a_0 - a_1 w_i)^2, \\ S_{\text{multiple}}(b_0, b_1, b_2, b_3) &= \sum_{i=1}^n (y_i - b_0 - b_1 w_i - b_2 h_i - b_3 t_i)^2. \end{aligned}$$

Let $(\hat{a}_0, \hat{a}_1)$ be a least-squares solution for the weight-only model. The coefficient vector $(\hat{a}_0, \hat{a}_1, 0, 0)$ is an available choice for the multiple regression model and satisfies

$$S_{\text{multiple}}(\hat{a}_0, \hat{a}_1, 0, 0) = S_{\text{weight}}(\hat{a}_0, \hat{a}_1).$$

Minimising over all four coefficients cannot give a larger objective value than this particular choice. Therefore,

$$\begin{aligned} \min_{(b_0, b_1, b_2, b_3) \in \mathbb{R}^4} S_{\text{multiple}}(b_0, b_1, b_2, b_3) &\leq S_{\text{multiple}}(\hat{a}_0, \hat{a}_1, 0, 0) \\ &= \min_{(a_0, a_1) \in \mathbb{R}^2} S_{\text{weight}}(a_0, a_1). \end{aligned}$$

This establishes the claimed inequality. $\square$

However, we should note that the inequality above need not be strict: additional predictors do not necessarily reduce the training sum of squared errors. Furthermore, the result concerns squared error on the training observations only. It does not guarantee an improvement in training absolute error or in performance on unseen observations.

For this reason, predictive performance was assessed using five-fold cross-validation, [5, Section 5.1.3], within the training set. The training observations were divided into five groups. Each model was fitted using four groups and evaluated on the remaining group, with this process repeated until every group had been used for validation. The same folds were used for all three approaches for a consistent comparison.

After cross-validation, each model was fitted using all 313 training observations and evaluated on the same 79 test observations. These test observations were excluded from coefficient

estimation and cross-validation. This separates the assessment of predictive performance from the observations used to fit the models.

3.2.4 Assumptions and interpretation

Fitting a least-squares model does not show that a linear relationship is appropriate for the data. This section sets out the assumptions used to interpret the models and explains what the residual plots can tell us.

Conditional mean specification. We can write a general regression model in the following way

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i,$$

and here we can interpret the linear expression as the conditional mean of fuel efficiency which assumes that

$$\mathbb{E}[\varepsilon_i \mid \mathbf{x}_i] = 0,$$

where $\mathbf{x}_i$ contains the predictor values for vehicle i . Equivalently,

$$\mathbb{E}[y_i \mid \mathbf{x}_i] = \beta_0 + \sum_{j=1}^p \beta_j x_{ij}.$$

This means that, for a given combination of predictor values, the unexplained component averages to zero in the population being modelled. Individual observations may still differ substantially from the conditional mean.

This assumption is not established merely by fitting a regression model. In particular, the fitted training residuals sum to zero when ordinary least squares includes an intercept, but this algebraic property does not prove that the conditional mean has been correctly specified. For background on the regression model and its error term, see [5, Sections 3.1–3.2].

Linearity and additivity. The models used in this investigation contain separate linear terms for the selected predictors, with no interaction or nonlinear terms. In the multiple regression model, a one-unit increase in predictor j changes predicted fuel efficiency by $\hat{\beta}_j$, holding the other predictors fixed.

Take for example, the predicted change associated with a one-pound increase in weight is the same at every horsepower and model year. A term such as w_i^2 would allow curvature in the relationship with weight, while a term such as $w_i h_i$ would allow the weight slope to depend on horsepower. Neither type of term was included in the fitted models. Interactions and nonlinear extensions of linear regression are discussed in [5, Section 3.3.2].

Error distribution and variability. Calculating the least-squares fit does not require normally distributed errors or constant error variance. Constant conditional error variance, which is commonly known as homoscedasticity, implies that

$$\text{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma^2$$

for all predictor values. The spread of the unexplained variation would remain constant across different types of vehicles.

Additional distributional and dependence assumptions are relevant to conventional coefficient

standard errors, confidence intervals and significance tests. For the associated inferential framework, see [5, Sections 3.1–3.2]. These procedures were not carried out in this investigation. Nevertheless, changing error variability remains relevant because it may indicate that predictions are less reliable for some vehicles than for others.

Residual assessment. A residual is the difference between an observed outcome and its fitted prediction:

$$r_i = y_i - \hat{y}_i.$$

Residuals are observable after fitting, whereas the theoretical errors ε_i depend on the unknown underlying model.

Plots of test residuals against predicted fuel efficiency were examined for systematic patterns. Curvature may suggest that the linear specification misses part of the relationship, while a changing vertical spread may indicate unequal error variability. These plots provide diagnostic evidence but do not prove that the model assumptions hold.

Generalisability and causal interpretation. Interpreting validation performance as an indication of performance on other vehicles requires those vehicles to be sufficiently comparable to the observed sample. Closely related observations appearing in both training and evaluation data could make prediction appear easier than it would be for substantially different vehicles.

The random split assesses prediction within the historical dataset; it does not directly assess forecasting for later model years. The results should therefore not be assumed to apply to modern vehicles or substantially different vehicle populations.

Finally, the fitted coefficients describe associations within the chosen models. Holding the other included predictors fixed is a mathematical interpretation of a coefficient, not evidence that changing a vehicle characteristic would cause the corresponding change in fuel efficiency.

3.3 Evaluation design

Three approaches were evaluated: a baseline that predicts the mean fuel efficiency of the training observations, a linear regression using vehicle weight alone, and a multiple linear regression using weight, horsepower and model year. The baseline provides a reference for assessing whether vehicle characteristics improve predictions beyond a single constant value.

3.3.1 Training and test split

The 392 complete observations were randomly divided into 313 training observations and 79 test observations, corresponding approximately to an 80:20 split. A random seed of 42 was used to make the split reproducible. All three approaches used the same training and test observations, ensuring that differences in performance were not due to different evaluation samples.

The split was random rather than chronological. Consequently, the evaluation assesses prediction for held-out vehicles from the historical dataset, rather than forecasting fuel efficiency for later model years.

3.3.2 Cross-validation and model comparison

Five-fold cross-validation was performed within the training set [5, Section 5.1.3]. The 313 training observations were shuffled and divided into five approximately equal groups, using a

random seed of 42. Each approach was fitted on four groups and evaluated on the remaining group. This was repeated five times so that each group served once as the validation set.

The same folds were used for all three approaches. For each fold, the regression coefficients were estimated using only that fold's training observations. Similarly, the baseline prediction was calculated from the mean outcome in those training observations, excluding the validation group.

Mean absolute error was calculated for each validation fold. The mean and standard deviation of these five errors were reported to describe average predictive performance and its variation across folds. The standard deviation is descriptive; it is not a confidence interval for predictive performance.

The scripts specify the predictor sets in advance of cross-validation and do not perform an automated search over alternative predictor combinations or tuning parameters. Cross-validation therefore compares the three specified approaches. It does not establish that the selected predictors are the best possible subset of the available variables.

After cross-validation, each approach was fitted to all 313 training observations and evaluated on the same 79 test observations. Test outcomes were used to calculate the final evaluation metrics, not to estimate coefficients or calculate cross-validation scores.

3.3.3 Evaluation metrics

Let m denote the number of observations being evaluated, y_i the observed fuel efficiency, and $\hat{y}_i$ its prediction. Three complementary metrics were calculated.

Mean absolute error. The mean absolute error measures the average magnitude of prediction errors:

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|. \quad (5)$$

It is expressed in miles per gallon, and smaller values indicate better predictive performance.

Root mean squared error. The root mean squared error is

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}. \quad (6)$$

It is also expressed in miles per gallon. Squaring the errors before averaging makes this metric more sensitive to large errors than mean absolute error. Smaller values indicate better performance.

Coefficient of determination. For an evaluation sample with nonzero outcome variation, the coefficient of determination is

$$R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y}_{\text{eval}})^2}, \quad \bar{y}_{\text{eval}} = \frac{1}{m} \sum_{i=1}^m y_i. \quad (7)$$

This metric is dimensionless, and larger values indicate better performance. A value of one represents perfect predictions. A value of zero indicates the same squared error as predicting the evaluation sample's mean, while a negative value indicates greater squared error than that reference.

The evaluation mean in Equation 7 is used only to calculate the metric. It differs from the training mean used by the fitted baseline. Consequently, the baseline's test-set R^2 is not necessarily zero.

Mean absolute error was used for cross-validation. All three metrics were reported for the test set, alongside plots of prediction errors to support interpretation.

4 Results

Table 3. Predictive performance on the same 79 test observations. Mean absolute error and root mean squared error are measured in miles per gallon; lower values indicate better performance. The coefficient of determination (R^2) is dimensionless; higher values indicate better performance.

Method	Mean absolute error	Root mean squared error	R^2
Mean baseline	5.881	7.185	-0.011
Weight regression	3.464	4.206	0.653
Multiple regression	2.509	3.255	0.792

Table 3 shows that the weight-only regression reduced test mean absolute error from 5.881 to 3.464 miles per gallon, an improvement of approximately 41.1% relative to the mean baseline. Its negative fitted weight coefficient indicates that heavier vehicles were associated with lower predicted fuel efficiency.

The multiple regression model reduced test mean absolute error further, to 2.509 miles per gallon. This represents an improvement of approximately 27.6% compared with the weight-only model. It also achieved the lowest mean cross-validation error, at 2.695 miles per gallon, compared with 3.317 for weight-only regression and 6.730 for the mean baseline.

The agreement between the cross-validation and test-set comparisons supports the conclusion that horsepower and model year jointly add predictive value beyond vehicle weight in this investigation. This comparison does not establish the separate contribution of each additional predictor or demonstrate causal effects on fuel efficiency.

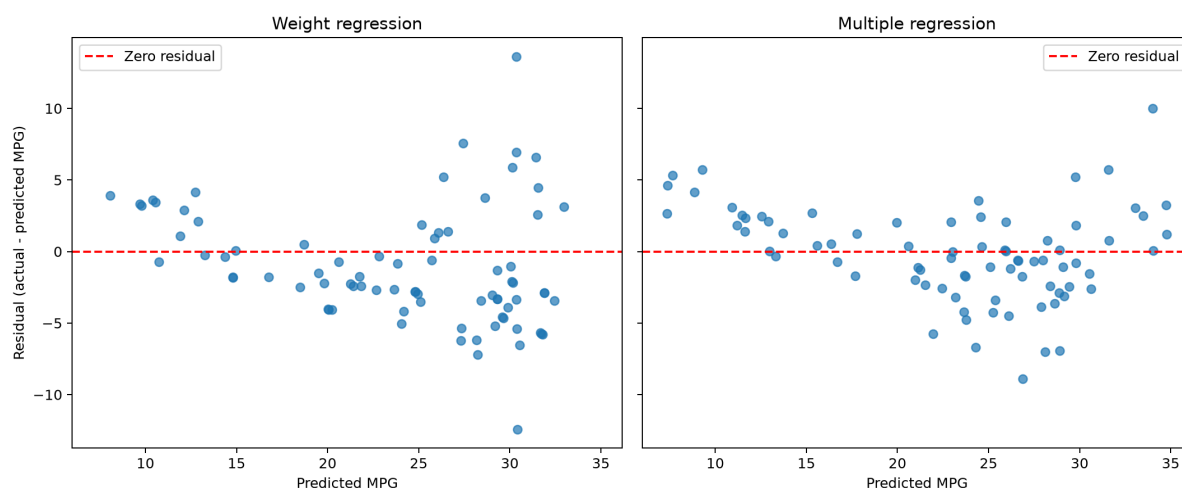


Figure 1. Test residuals against predicted fuel efficiency for the weight-only and multiple regression models. Both panels use the same axis scales. Residuals are observed minus predicted fuel efficiency, measured in miles per gallon. Positive values indicate underprediction, and negative values indicate overprediction. The red dashed line marks zero prediction error.

Figure 1 illustrates the prediction errors underlying the numerical comparison. The weight-only model shows substantial variation in residuals at higher predicted fuel efficiency. Although the

multiple regression model reduces overall prediction error, visible structure remains: residuals tend to be positive at low predicted values and negative across much of the middle range, with several positive residuals at the upper end. This pattern suggests that the additive linear specification may not fully capture the relationship between the predictors and fuel efficiency.

5 Discussion

Weight alone provided useful predictions, but including horsepower and model year reduced errors further in both cross-validation and the test set. This suggests that vehicles with similar weights differ in ways that matter for predicting fuel efficiency. Since horsepower and model year were added together, this comparison does not show how much each contributes separately.

Horsepower and model year may capture differences in engine characteristics and vehicle design that weight alone misses. This offers a possible explanation for the improvement, although the analysis does not establish which mechanisms are responsible.

The multiple regression model missed the observed test values by 2.5 miles per gallon on average in absolute terms, although some errors were much larger. No acceptable error threshold was defined for this project, so this result supports a comparison between models rather than a claim that the model is ready for practical use.

The lower average error does not mean that the multiple regression model captured the relationship fully. Patterns remain in the residuals, suggesting that some prediction errors are systematic. These patterns give a reason to explore other model specifications [6].

5.1 Diagnostics and robustness

The residual plots show that prediction errors were not uniformly distributed across the range of predicted fuel efficiency. For the weight-only model, residuals were generally positive at low predicted values, while a wider spread appeared at higher predicted values. This indicates that the model underpredicted fuel efficiency for several vehicles at the lower end of its prediction range and produced substantial errors in both directions at the upper end.

The multiple regression model reduced overall error, but systematic patterns remained. Residuals were generally positive at low predicted values and negative across much of the middle range, with several positive residuals at the upper end. This suggests that the additive linear specification may not fully represent the relationship between the selected predictors and fuel efficiency. Curvature, interactions or omitted vehicle characteristics are possible explanations, but the residual plots alone cannot distinguish between them [6].

Several test observations had comparatively large residuals. These observations were retained: a large prediction error does not itself establish that an observation is incorrect. No formal influence analysis was performed, so the investigation does not establish whether particular training observations had a disproportionate effect on the fitted coefficients.

Five-fold cross-validation provided a limited assessment of sensitivity to the training observations used for fitting. The multiple regression model achieved a mean validation absolute error of 2.695 miles per gallon, with a standard deviation across folds of 0.243. The corresponding values were 3.317 and 0.445 for weight-only regression, and 6.730 and 0.747 for the mean baseline.

The lower average error and smaller variation across folds support the multiple regression model within this evaluation. However, the fold standard deviations are descriptive and do not constitute confidence intervals or a formal test of differences between models. No repeated train-test splits, alternative missing-value treatments or alternative predictor specifications were evaluated. Robustness to those choices therefore remains untested.

5.2 Limitations

The analysis used 392 complete observations, of which 79 were reserved for testing. Performance estimates therefore depend on a relatively small test sample and one random train-test split. Cross-validation provides additional evidence from the training data, but does not remove uncertainty about performance on other samples.

Six observations were excluded because horsepower values were missing. Using the same complete observations allowed consistent comparisons between models, but also removed vehicles that could otherwise have been used in the weight-only analysis. If the excluded vehicles differed systematically from those retained, complete-case analysis could affect the representativeness of the findings.

I explored the full cleaned dataset before fitting the models. If those observations influenced my choice of predictors, the test data indirectly informed model development. The test set was excluded from fitting, but was not reserved before all exploratory work.

Only two regression specifications and a constant baseline were compared. The multiple regression model included weight, horsepower and model year, but omitted other recorded characteristics and contained no nonlinear or interaction terms. Its superior performance among the evaluated approaches does not establish that it is the best available model. Furthermore, adding horsepower and model year together does not identify their separate predictive contributions.

The remaining residual patterns indicate that some systematic variation may be unexplained. No formal tests of error variance, influence diagnostics or coefficient confidence intervals were calculated. Consequently, the investigation provides a predictive comparison rather than a comprehensive assessment of statistical inference assumptions.

Finally, the dataset represents vehicles from model years 1970 to 1982. A random split evaluates prediction within this historical sample, rather than forecasting for later vehicle generations. The findings should not be assumed to generalise to modern vehicles.

6 Conclusion

Linear regression provided useful predictions of fuel efficiency in the Auto MPG dataset. Weight-only regression reduced test mean absolute error by 41.1% compared with the mean baseline. Adding horsepower and model year reduced the error by a further 27.6%, to 2.509 miles per gallon, and also gave the lowest mean cross-validation error.

Nevertheless, systematic residual patterns remained, and the evaluation was limited to a small historical dataset. The multiple regression model was the strongest of the three approaches tested, but its performance should not be assumed to extend to modern vehicles or interpreted as evidence of causal relationships.

6.1 Future work

A first extension would investigate nonlinear and interaction terms, guided by the residual patterns. For example, a squared weight term could represent curvature, while an interaction between weight and horsepower could allow their predictive relationship to vary jointly. These extensions should be compared using validation data to determine whether the additional flexibility improves prediction.

A second step would assess sensitivity to modelling choices. Comparing models that add horsepower and model year separately would help distinguish their predictive contributions. Alternative treatments of missing horsepower values could also show whether excluding incomplete observations materially affects the findings. Any estimated replacement values should be learned within each training fold to avoid using validation information.

Finally, further model development should use a clearly separated evaluation procedure. Repeated or nested cross-validation could assess sensitivity to data partitions and account for model selection. A new, untouched dataset would provide stronger evidence of generalisation; evaluating on more recent vehicles would specifically test whether the relationships extend beyond the historical sample.

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